

COCO AI

Correspondence Operations Intelligence Platform

A single place to capture, own, measure and instantly retrieve every piece of correspondence the organisation sends or receives — with AI doing the reading, the routing and the remembering. Searchable. Auditable. Actionable.

Prepared for: COCO (Corporate Correspondence)

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Reference prototype: Working application — nine modules, sample corpus spanning 2003–2026

1. Overview

COCO sits at the centre of how the organisation talks to the outside world. Every letter from a regulator, every notice from a ministry, every request from a vendor, every concern raised by a host community, and every formal reply we send back — all of it passes through this function. Today that flow is run almost entirely out of email and personal spreadsheets. It works, in the sense that letters do eventually get answered, but it is impossible to see, impossible to measure, and very hard to audit.

COCO AI replaces that scattered way of working with one system. The moment a letter arrives it is logged, given an owner and a deadline, and tracked all the way through to dispatch. Nothing sits in an inbox unseen. Leadership gets a live picture of the whole operation instead of a monthly guess. And because every record is stored and indexed, a letter from 2003 can be found in seconds rather than days.

On top of that foundation sits the part people remember from the demo: the AI reads incoming documents, pulls out the sender, the reference, the deadline, the required action and the risk, and writes a short summary — turning an hour of manual reading and typing into a few seconds of review. A working prototype already demonstrates every capability described in this document. What follows is what it must do to become a production system.

2. The problem we are solving

There is a short list of questions that leadership simply cannot answer today, and every one of them matters:

- Where is a particular correspondence right now, and who owns it?
- When did we actually receive it, and how long has it been sitting?
- What is due, what is overdue, and who or what is causing the delay?
- What business value — an approval, a permit, a payment — is being blocked while we wait?
- Can we put our hands on a specific letter from years ago, instantly, with proof of when it arrived?

The reasons are structural, not personal. Correspondence lives in individual mailboxes. There is no single register of record. Ownership, deadlines and service levels are tracked informally if at all. The historical archive is not searchable. And every incoming letter is read and re-keyed by hand. The cost shows up as missed statutory deadlines, slow responses to government and communities, regulatory exposure, lost institutional memory, and a leadership team operating without visibility.

3. Goals

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|-----------|--|
| G1 | Establish a single register of record for all incoming and outgoing correspondence. |
| G2 | Give every record a complete, timestamped, tamper-evident audit trail. |
| G3 | Turn every correspondence into a tracked action with an owner, a due date and a service level. |
| G4 | Make any record — back to 2003 — retrievable in seconds. |
| G5 | Use AI to extract key fields, assess risk and summarise, with a person always in the loop. |
| G6 | Give leadership a live, honest view of volume, service levels, bottlenecks and team performance. |

4. Success criteria

We will consider the production system a success when the following are true in day-to-day use — not in a demo, but as the normal way COCO works. These are the conditions the delivery will be judged against, and they double as the high-level acceptance criteria.

- **Nothing is invisible.** Every piece of correspondence in or out is in the system, each with a named owner and a due date. There is no parallel tracking happening in private spreadsheets.
- **The questions in Section 2 are answerable on the spot.** Any manager can stand in front of the dashboard and say where something is, who has it, and whether it is late — without phoning around.
- **History is genuinely retrievable.** A named officer can find any historical letter, including one from 2003, in under ten seconds, and can show when it was received and how it was handled.
- **The AI saves real time.** An incoming letter can be turned into a structured, owned record from an uploaded document with a fraction of the manual typing it takes today, and the person reviewing trusts the result enough to use it.
- **Service levels are met and seen to be met.** SLA compliance holds at or above target, and when something is about to breach, the team sees it early enough to act.
- **It stands up to an audit.** An auditor can be handed a record and shown a complete, defensible trail from receipt to closure, with nothing edited away silently.
- **People actually use it.** Officers prefer it to the old way because it removes drudgery, and adoption does not have to be forced.

The measurable targets that sit underneath these criteria are in Section 5.

5. Success metrics

Concrete numbers we can track. The baselines come from the sample corpus and current working assumptions; they should be confirmed against live data early in delivery.

ID	WHAT WE MEASURE	BASELINE	TARGET
M1	Average assignment (transit) time, received to assigned	~5.2 days	2 days or less
M2	SLA compliance	~86%	90%+ sustained
M3	Overdue open actions	Largely unmeasured	Down 50% within two quarters
M4	Time to retrieve a historical record	Hours to days	Under 10 seconds
M5	Manual data entry per incoming letter	Fully manual	Down ~80% via AI extraction
M6	Correspondence with a named owner and due date	Partial	100%
M7	Records with a complete audit trail	Inconsistent	100%

6. Who uses it

ROLE	WHAT THEY NEED FROM COCO AI
COCO Officer	Log, own, action and dispatch correspondence; meet deadlines without chasing.
COCO Manager	See the whole pipeline; manage service levels, bottlenecks and team workload.
Executive / Leadership	Transparency and a clear narrative; assurance that risk is under control.
Department Reviewer	Receive routed items, contribute input and approve drafts.
Auditor / Compliance	Prove receipt dates and follow a complete trail end to end.
Records / Archivist	Retrieve and curate the historical corpus.
Administrator	Manage access, roles and the underlying integrations.

7. User stories

Written from the point of view of the people who will live in this system every day. Each one carries an implied test: it is done when the person described can do the thing described, on real data, without a workaround.

Capturing and owning correspondence

As a COCO Officer, I want every incoming letter logged the moment it arrives, with a reference, sender, date received and date stamped, so that nothing slips into a mailbox and disappears.

Done when: a new incoming record can be created in under a minute and immediately appears in the register with its transit clock running.

As a COCO Manager, I want each correspondence assigned to a named owner with a due date and priority, so that responsibility is never ambiguous.

Done when: no item can sit in the pipeline without an owner, and unassigned items are visibly flagged.

As a COCO Officer, I want the system to calculate transit time and COCO processing time for me, so that the delays we have always argued about are finally measured, not guessed.

Done when: both figures appear automatically from the recorded dates and feed the dashboards.

Staying on top of the work

As a COCO Officer, I want overdue and due-soon items to stand out at a glance, so that I work on the right thing first.

Done when: the Action Tracker visibly distinguishes overdue, due-soon and on-track, and I can filter to just my overdue items.

As a COCO Manager, I want to see which departments and which officers are slipping on service levels, so that I can step in before a breach becomes a problem.

Done when: SLA compliance and overdue load are shown per team and per officer, and a looming breach is visible early.

Reading and routing with AI

As a COCO Officer, I want to upload a scanned letter and have the system pull out the sender, reference, due date, required action and risk, then write a summary, so that I spend my time deciding, not typing.

Done when: an uploaded document produces the extracted fields with a confidence score, and I can turn the result into a draft record with one step.

As a COCO Manager, I want the AI to flag the risk level of an incoming item, so that a critical regulatory deadline is never treated as routine.

Done when: each AI-processed item carries a risk level that a human can confirm or override.

Finding anything, proving everything

As a Records Archivist, I want to search the entire history by words, sender, year, department or theme, so that retrieving a decades-old letter takes seconds.

Done when: a free-text query returns ranked results across 2003 to today, narrowable by every listed filter.

As an Auditor, I want to open any record and see a complete timeline from receipt to closure, so that I can verify how and when we handled it.

Done when: the record profile shows full metadata and an audit trail, and any change is recorded rather than overwritten.

Seeing the whole picture

As an Executive, I want a dashboard of volume, open and overdue actions, service levels and bottlenecks, plus a short written summary of what is happening, so that I understand the state of correspondence without commissioning a report.

Done when: leadership can read the current position and the key shifts (for example, “regulatory correspondence up 18% this quarter”) directly from the screen.

8. Functional requirements

MUST = required for the first production release. SHOULD = strongly wanted. MAY = optional.

8.1 Executive Dashboard

- **DASH-01 (MUST)** — Headline figures: incoming and outgoing totals, open actions, overdue actions, SLA compliance, average response time, average transit time, average COCO processing time, critical-open count.
- **DASH-02 (MUST)** — Volume trend of incoming versus outgoing over a rolling period.
- **DASH-03 (MUST)** — Department performance: volume, overdue load and SLA.
- **DASH-04 (MUST)** — Top bottlenecks: the oldest open items, with owner and age, click-through to the record.
- **DASH-05 (SHOULD)** — Stakeholder mix and pipeline-status breakdown.
- **DASH-06 (SHOULD)** — A panel of AI executive insights linking to full reporting.

8.2 Incoming Correspondence

- **IN-01 (MUST)** — Capture reference number, subject, sender, stakeholder type, date received, date stamped, date assigned, assigned owner, department, due date, priority, status and document type.
- **IN-02 (MUST)** — Automatically calculate *transit time* (date assigned minus date stamped).
- **IN-03 (MUST)** — Filter by department, status, priority and stakeholder type, plus free-text search across reference, subject, sender and owner.
- **IN-04 (MUST)** — Clearly flag overdue items, and open a full record on selection.
- **IN-05 (SHOULD)** — Export the filtered register, and log a new item manually or straight from an AI extraction.

8.3 Outgoing Correspondence

- **OUT-01 (MUST)** — Capture reference, subject, recipient, stakeholder type, department, draft-completed date, submitted-to-COCO date, processed date, sent date, delivery status, document type and owner.
- **OUT-02 (MUST)** — Automatically calculate *COCO processing time* (processed date minus submitted-to-COCO date).
- **OUT-03 (MUST)** — Track the delivery lifecycle from draft through to delivered, acknowledged or returned, with filtering and detail equivalent to Incoming.

8.4 Action Tracker

- **ACT-01 (MUST)** — Every correspondence produces a trackable action carrying reference, description, owner, department, due date, priority, status, progress and direction.
- **ACT-02 (MUST)** — Support the lifecycle: Received, Assigned, In Progress, Response Drafted, Submitted To COCO, Sent, Closed.
- **ACT-03 (MUST)** — Reminder indicators for overdue, due-soon and on-track, with counters that filter the list.
- **ACT-04 (SHOULD)** — Filter by owner, department, status and direction.

8.5 SLA Monitor

- **SLA-01 (MUST)** — Overall compliance against target, overdue actions and average handling time.

- **SLA-02 (MUST)** — Team and department comparison, and a compliance trend over time against target.
- **SLA-03 (SHOULD)** — Configurable SLA windows by priority and stakeholder type.

8.6 Performance Dashboard

- **PERF-01 (MUST)** — Per-officer metrics: letters processed, average turnaround, SLA compliance, overdue count and open workload.
- **PERF-02 (MUST)** — A leaderboard ranked by service level then turnaround.
- **PERF-03 (SHOULD)** — Compare officers across volume, speed and workload, and highlight overload for rebalancing.

8.7 AI Letter Intelligence

- **AI-01 (MUST)** — An upload area (drag-and-drop and browse) accepting PDF and common image formats.
- **AI-02 (MUST)** — Extract sender, recipient, subject, reference number, due date, required action, stakeholder type and risk level.
- **AI-03 (MUST)** — Produce a written summary and show an extraction confidence score.
- **AI-04 (MUST)** — Let the result become a draft record or action, confirmed by a person. The AI never sends anything on its own.
- **AI-05 (SHOULD)** — Show the processing stages so the user can see what the system is doing.

8.8 Executive Reporting

- **REP-01 (MUST)** — Generate plain-language insights with direction and magnitude, for example a quarter-on-quarter rise in regulatory correspondence or a fall in assignment time.
- **REP-02 (MUST)** — A management summary with supporting charts: volume, status mix, historical volume by year and stakeholder mix.
- **REP-03 (SHOULD)** — Export an executive brief, and optionally schedule its delivery.

8.9 Historical Search & Retrieval — the flagship

- **SEARCH-01 (MUST)** — Full-text search across all records from 2003 to the present, over subject, sender, recipient, reference, summary and tags.
- **SEARCH-02 (MUST)** — Filters for year, sender, recipient, department, stakeholder type, document type and status.
- **SEARCH-03 (MUST)** — Results as cards showing title, date, sender, recipient, summary, department, tags and direction.
- **SEARCH-04 (MUST)** — An AI summary panel that synthesises the current result set: count, date range, dominant stakeholder and recurring themes.
- **SEARCH-05 (MUST)** — Selecting a result opens a full profile with metadata, summary, audit timeline, tags and related correspondence.
- **SEARCH-06 (SHOULD)** — Semantic search alongside keyword search, so a concept matches even when the words differ.

9. Data model

The core entities below come straight from the prototype's typed model and map cleanly onto Dataverse tables.

- **Incoming Correspondence** — reference, subject, sender, stakeholder type, dates (received, stamped, assigned, due), owner, department, priority, status, document type, derived transit time, tags, summary.
- **Outgoing Correspondence** — reference, subject, recipient, stakeholder type, department, dates (draft completed, submitted to COCO, processed, sent), delivery status, document type, owner, derived COCO processing time, tags, summary.
- **Action Item** — correspondence reference, description, owner, department, due date, priority, status, progress, direction.
- **Officer** — name, role, department, letters processed, average turnaround, SLA compliance, overdue count, open workload.
- **Archive Record** — reference, title, date, sender, recipient, direction, department, stakeholder type, document type, status, summary, tags, page/size metadata, and a link to the source document in SharePoint.

Shared enumerations cover stakeholder type, priority, risk level, department, document type, action status and delivery status. Derived values — transit time, COCO processing time, the overdue flag, action age and SLA compliance — are calculated in one place so every screen agrees on the numbers.

10. AI capability and governance

The AI work runs on Azure OpenAI inside the organisation's own tenant. It performs document reading and layout analysis, classification of document and stakeholder type, structured field extraction, risk scoring, summarisation, and the embeddings that power semantic historical search.

The governing principles are non-negotiable:

- A person reviews and approves every output that has an effect. The AI never dispatches correspondence.
- Confidence is always shown; low-confidence results are flagged for human attention.
- Documents and prompts are processed in the chosen region and are not used to train any public model.
- AI-suggested values are stored alongside the human-confirmed values, and each extracted field links back to its place in the source document.

11. How it connects to our systems

The prototype is deliberately built around a clean seam: the interface talks to a data and analytics layer, never directly to storage. In production each connector below replaces the sample data with a live service, and the interface does not have to change.

SERVICE	ITS JOB IN COCO AI
Microsoft 365 / Entra ID	Single sign-on, identity and role assignment.

SERVICE	ITS JOB IN COCO AI
Outlook / Exchange	Pull incoming correspondence from monitored COCO mailboxes; send approved replies.
SharePoint	The document library of record for scanned originals, with retention back to 2003.
Dataverse	The system of record for correspondence, actions, officers, SLAs and audit history.
Power Automate	Routing, reminders, escalations and SLA-breach workflows when records change.
Azure OpenAI	Document understanding, extraction, summarisation and semantic search.
Azure AI Search	A hybrid keyword and meaning-based index across the full archive.

Worth stating plainly: Dataverse and Power Automate can be the backend and workflow engine, while COCO AI provides the experience layer — the dashboards, the AI intelligence and the search. These are complementary, not competing, choices.

12. Non-functional requirements

- **Security.** Access through Entra ID single sign-on, with role-based control for officer, manager, executive, auditor and administrator. Encryption in transit and at rest. Access is logged. Sensitive items can be restricted by department or classification.
- **Auditability.** A timestamped trail per record from receipt to closure. Corrections are tracked edits, never silent overwrites.
- **Privacy and residency.** Configurable data residency and compliance with applicable data-protection obligations.
- **Performance.** Search should return in under two seconds across the full archive; everyday interactions should feel instant.
- **Availability.** Run on managed cloud with high availability; source documents remain independently safe in SharePoint.
- **Accessibility and reach.** Aim for WCAG 2.1 AA; work responsively on desktop, tablet and phone.
- **Scale and retention.** Hold twenty-plus years of records and a growing annual volume without redesign; govern retention and deletion by policy, with deletions logged.

13. Bringing the history in (2003–2026)

This is the largest single effort in the programme, and it is honest to say so up front. It is also largely independent of which front-end technology is chosen, so it should be planned and resourced as its own workstream.

1. Inventory the existing digital archives and the physical records.
2. Import structured and digital records into Dataverse; place originals in SharePoint.
3. Scan physical correspondence in batches, with OCR and AI auto-tagging on the way in.
4. Reconcile and quality-check; send low-confidence extractions for human review.
5. Index the corpus for fast retrieval.
6. Phase by value: load the periods and regulators that get referenced most often first, then backfill the rest.

14. Delivery roadmap

PHASE	WHAT IT DELIVERS
0 — Foundation (done)	The working prototype: all nine modules on a sample corpus, validating the experience, the data model and the AI workflow.
1 — Identity & records	Live single sign-on and Dataverse as the system of record, with role-based access and the audit trail.
2 — Ingestion	Automatic capture from Outlook and the SharePoint document store.
3 — Intelligence	Live Azure OpenAI extraction and summaries, replacing the simulated version, with confidence and human review.
4 — Search at scale	Hybrid keyword and semantic search across the whole archive.
5 — Workflow & alerts	Routing, reminders and SLA escalations through Power Automate.
6 — Migration	The historical corpus loaded, phased by value.
7 — Hardening & rollout	Performance, accessibility, training, and a pilot that scales to full use.

15. Risks and how we handle them

RISK	HOW WE MANAGE IT
The historical migration is bigger than expected	Treat it as a dedicated, phased workstream; prioritise by how often records are actually needed.
AI extraction makes mistakes	Keep a person in the loop, set confidence thresholds and link every field back to the source.
People do not adopt it	It removes manual toil, so it earns its place; pilot one department and keep the experience familiar.
Sensitive data is over-exposed	Role-based access, classification, encryption and access logging.

RISK	HOW WE MANAGE IT
Scope creeps beyond COCO	Hold the scope; defer everything else to a future-candidates list.
The custom-versus-low-code debate stalls the decision	Frame it as backend versus experience layer and decide that explicitly, rather than as an either/or.

16. How the prototype is actually built

This section exists so that anyone assessing the prototype can judge it as an engineering foundation rather than a throwaway mock-up. The structure was chosen deliberately, with production in mind from the start.

- **Layered, with a swappable storage seam.** The interface reads from an analytics layer, which reads from a data interface, which today points at sample data and tomorrow points at Dataverse. The interface never touches raw storage, so the backend can be changed without rewriting the screens.
- **A single typed domain model** shared across the data, analytics and interface layers — the same shape described in Section 9, so the move to Dataverse is a mapping exercise, not a redesign.
- **An explicit integration layer** that already names the contracts for Microsoft 365, Outlook, SharePoint, Dataverse, Power Automate and Azure OpenAI — the precise points where live services attach.
- **Reproducible, seeded data** so the system renders consistently and demonstrations stay stable.
- **A disciplined server-and-client split**, rendering on the server by default and using interactivity only where it is needed, with a type-checked, clean build across roughly 492 pre-rendered pages.

The stack is Next.js with the App Router, React, TypeScript, Tailwind CSS, a shadcn/ui component base, Recharts for the dashboards and Lucide for iconography. The prototype is hosted on managed cloud; production would sit in Azure alongside everything else.

17. Custom application or Power Platform?

A balanced position, so the tooling question does not derail the decision:

- Power Apps, Power Automate and Dataverse are excellent for forms, workflow and a governed, low-maintenance data backend. They make a strong system of record and workflow engine.
- A custom experience layer is stronger exactly where low-code is constrained: bespoke AI document intelligence, a rich search experience with an AI summary panel, and executive-grade dashboards.
- The sensible model is to use Dataverse and Power Automate as the backend, and COCO AI as the front-end experience. That protects any Power Platform investment while keeping the AI and search capabilities that set this apart in review.
- The real decision to make is who owns the backend versus the experience layer — not a binary choice between the two tools.

18. Developer hand-off notes

A short orientation for whoever picks the code up next. The goal of the build was to make the path to production a matter of connecting services, not rewriting the application.

Where it lives and how to run it

- The codebase is a single Next.js (App Router) project in TypeScript. Standard workflow: `npm install`, then `npm run dev` for local development, or `npm run build` and `npm start` for a production build. It builds clean and lint-clean.
- It runs anywhere Node runs; the prototype is hosted on Vercel, and production is intended for Azure.

How the project is organised

- `app/` — one folder per module (dashboard, incoming, outgoing, actions, sla, performance, intelligence, reporting, search, plus the record profile under search).
- `components/` — the interface: shared UI primitives, layout, charts, the interactive registers, the AI experience and the search experience.
- `lib/` — the brains: the typed domain model, the data layer, the analytics aggregations, the AI extraction module and the integration contracts.

The one thing to understand first

Everything flows through a single data seam. The screens never read raw data; they read from the analytics layer, which reads from the data layer. Today that data layer returns deterministic sample records. To go to production, replace that layer with calls to Dataverse (and SharePoint for documents) while keeping the same return shapes — the domain types in `lib` are written to match the intended Dataverse tables. The screens should not need to change.

What is real and what is simulated

- **Real:** all nine module interfaces, the filtering and search behaviour, the derived calculations (transit time, COCO processing time, SLA, overdue, age), the dashboards and the record profiles.
- **Simulated, by design:** the data itself, and the AI extraction (it returns realistic structured results rather than calling a live model). The integration layer is defined as contracts, not yet wired to live services.

The recommended order of work

1. Stand up identity (Entra ID single sign-on) and the Dataverse tables from Section 9.
2. Point the data layer at Dataverse, table by table, leaving the interface untouched.
3. Wire ingestion from Outlook and document storage in SharePoint.
4. Replace the simulated extraction with a live Azure OpenAI call, preserving the confidence-and-review pattern.
5. Add Azure AI Search behind the historical search screen for scale and semantic matching.
6. Layer Power Automate flows on top for routing, reminders and escalations.

In short: the experience, the data shapes and the workflow logic are already settled. The remaining work is integration and migration, both of which the structure was built to accommodate.

19. Glossary

- **COCO** — the Corporate Correspondence function.
- **Transit time** — days from stamping an incoming letter to assigning it.
- **COCO processing time** — days from submitting an outgoing letter to COCO until it is processed.
- **SLA** — service level agreement; the target turnaround window.
- **Stakeholder type** — the category of external party, such as regulator, community or vendor.
- **System of record** — the authoritative store for a piece of data; here, Dataverse.
- **Human in the loop** — a person reviews and approves AI output before it takes effect.